Faculty of Computing

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SECR1013 Digital Logic

Project Report

Section 02

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Group 3

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Dedication and Acknowledgement

We would like to sincerely thank our lecturer, Dr. Zuriahati, for her patience, mentorship, and vital guidance during the elevator controller system's development. Her opinions and thoughts have greatly helped to shape the project and improve its quality.

We also thank our team members for their involvement and assistance during the process. Their competence and willingness to share knowledge allowed us to accomplish this project and deliver it in the most effective method possible.

Furthermore, we would like to thank the University of Technology Malaysia for providing the resources and facilities that allowed us to complete this project effectively.

Background

The project at hand involves the application of knowledge gained from a dedicated course, where a collaborative effort by a group of 3 to 4 students is underway to simulate an imaginary case study. The primary objective is the design and implementation of an electronic controller tailored for an elevator/lift system within a hotel building.

In the envisioned scenario, a hotel visitor engages with the elevator by inputting their desired level, tagging their room card, and initiating the UP/DOWN command. The elevator system then seamlessly navigates through the building, counting levels until it reaches the specified floor. This project not only serves as a practical application of learned concepts but also provides an opportunity for group exploration, creative problem-solving, and effective reporting and presentation.

The core design requirements for the electronic controller include managing a 8-floor elevator system, utilising a 3-bit up/down counter, and operating in synchronous mode through the implementation of JK flip-flops. The success of this project holds significance in showcasing the integration of theoretical knowledge into a practical application, addressing the vital role of efficient elevator systems in multi-story buildings, particularly within the context of a hotel environment.

As the project progresses, the focus will shift towards detailed design, implementation, and testing phases, with the ultimate goal of delivering a functional electronic controller that meets the simulated needs of the case study.

Problem Statement

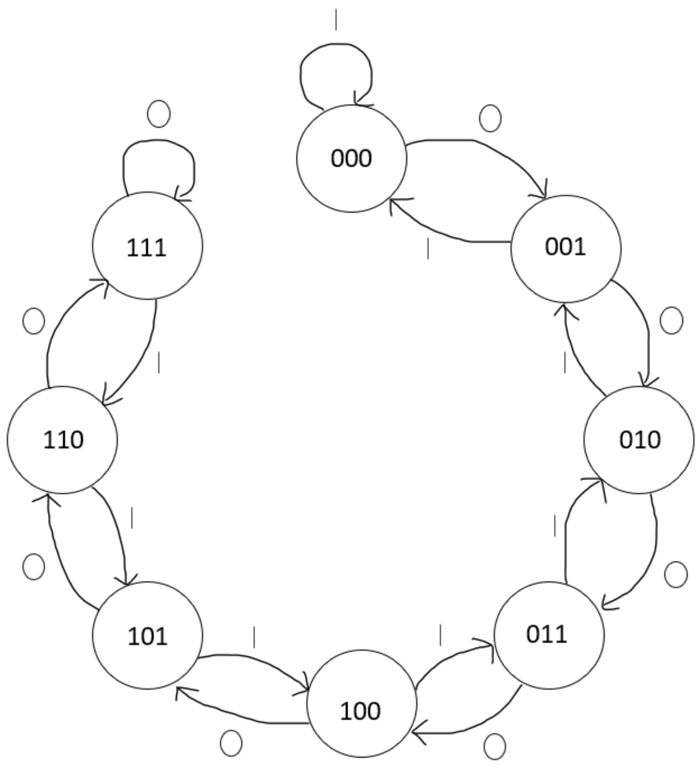
In the context of designing an electronic controller for an elevator/lift system within a hotel building, the project faces several key challenges and problem areas that necessitate thoughtful consideration and innovative solutions.

1. User Interaction Complexity: Streamlining the multi-step interaction process, encompassing level input, card tagging, and UP/DOWN initiation, for enhanced user-friendliness.
2. Synchronisation and Counting Mechanism: Ensuring precise synchronisation and accurate counting in the 8-floor elevator system with a 3-bit up/down counter, operating in synchronous mode.
3. Integration of JK Flip-Flops: Seamless integration of JK flip-flops as fundamental components within the electronic controller, addressing potential pitfalls and ensuring reliability.
4. Error Handling and Recovery: Implementing robust error detection and recovery mechanisms for scenarios like incorrect floor inputs or card malfunctions.

Suggested Solution

Our project is an 8 floor elevator system including the ground floor. We use a 3-bit up/down counter with 0 = count UP and 1 = count DOWN. This elevator functions in synchronous mode and uses a JK flip flop.

- **State diagram**



When the input X is 0, the elevator will go up by counting up from one level to another until it reaches the desired level. The highest level is level 7 so it cannot go up furthermore. Same goes to input X is 1, the elevator will go down by counting down. The lowest level is the ground floor so it cannot go down any further. This forms a saturated counter as it repeats the maximum count if count up or repeat the minimum if count down.

- **Next State and flip flop transition table**

Input	Present State			Next State			JK Flip Flop					
X	Q2	Q1	Q0	Q2+	Q1+	Q0+	J2	K2	J1	K1	J0	K0
0	0	0	0	0	0	1	0	X	0	X	1	X
0	0	0	1	0	1	0	0	X	1	X	X	1
0	0	1	0	0	1	1	0	X	X	0	1	X
0	0	1	1	1	0	0	1	X	X	1	X	1
0	1	0	0	1	0	1	X	0	0	X	1	X
0	1	0	1	1	1	0	X	0	1	X	X	1
0	1	1	0	1	1	1	X	0	X	0	1	X
0	1	1	1	1	1	1	X	0	X	0	X	0
1	0	0	0	0	0	0	0	X	0	X	0	X
1	0	0	1	0	0	0	0	X	0	X	X	1
1	0	1	0	0	0	1	0	X	X	1	1	X
1	0	1	1	0	1	0	0	X	X	0	X	1
1	1	0	0	0	1	1	X	1	1	X	1	X
1	1	0	1	1	0	0	X	0	0	X	X	1
1	1	1	0	1	0	1	X	0	X	1	1	X
1	1	1	1	1	1	0	X	0	X	0	X	1

We filled in the next state by following the state diagram. For the flip flop transition table, we followed the FF excitation table to fill in. Below is the JK FF excitation table.

Present State	Next State	FF State	
Qn	Qn+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

- **K-map**

XQ2 \ Q1Q0	00	01	11	10
00	0	0	1	0
01	X	X	X	X
11	X	X	X	X
10	0	0	0	0

$J2 = X'Q_1Q_0$

XQ2 \ Q1Q0	00	01	11	10
00	X	X	X	X
01	0	0	0	0
11	1	0	0	0
10	X	X	X	X

$K2 = XQ_1'Q_0'$

XQ2 \ Q1Q0	00	01	11	10
00	0	1	X	X
01	0	1	X	X
11	1	0	X	X
10	0	0	X	X

$J1 = X'Q_0 + XQ_2Q_0'$

XQ2 \ Q1Q0	00	01	11	10
00	X	X	1	0
01	X	X	0	0
11	X	X	0	1
10	X	X	0	1

$K1 = XQ_0' + X'Q_2'Q_0$

XQ2 \ Q1Q0	00	01	11	10
00	1	X	X	1
01	1	X	X	1
11	1	X	X	1
10	0	X	X	1

$J0 = X' + Q_2 + Q_1$

XQ2 \ Q1Q0	00	01	11	10
00	X	1	1	X
01	X	1	0	X
11	X	1	1	X
10	X	1	1	X

$KO = Q_1' + X + Q_2'$

- Combinational circuit component

- 4 bit decoder
- 4 bit comparator
- Basic gates
- Switch (power on current floor LED)
- Button (clock for lift move)
- LED + 8 bit output LED arrays
- 7 segment display
- Hex digit input

- **Sequential circuit component**

- 3 bit up and down counter (JK flip flop)
- Clock disabler

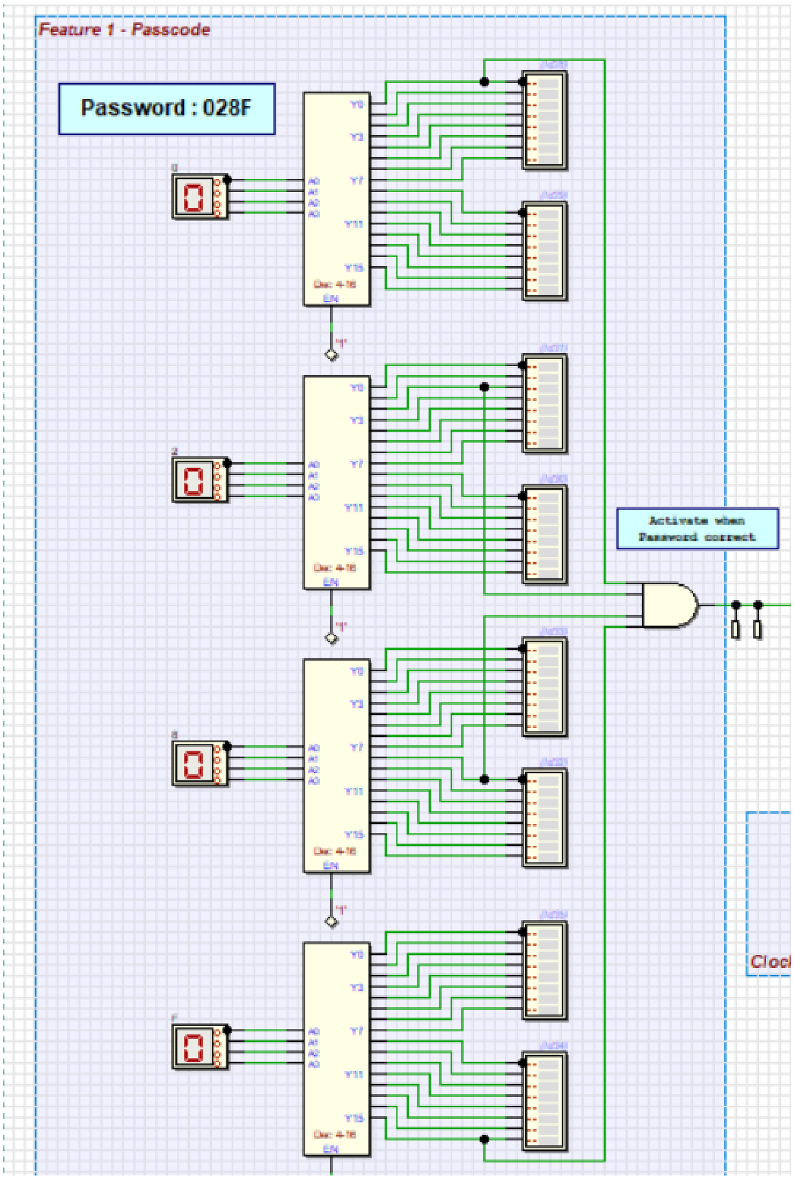
- Enhanced features

- 4-digit passcode system
- Detect card system
- Auto input X (count up and count down)

System Implementation

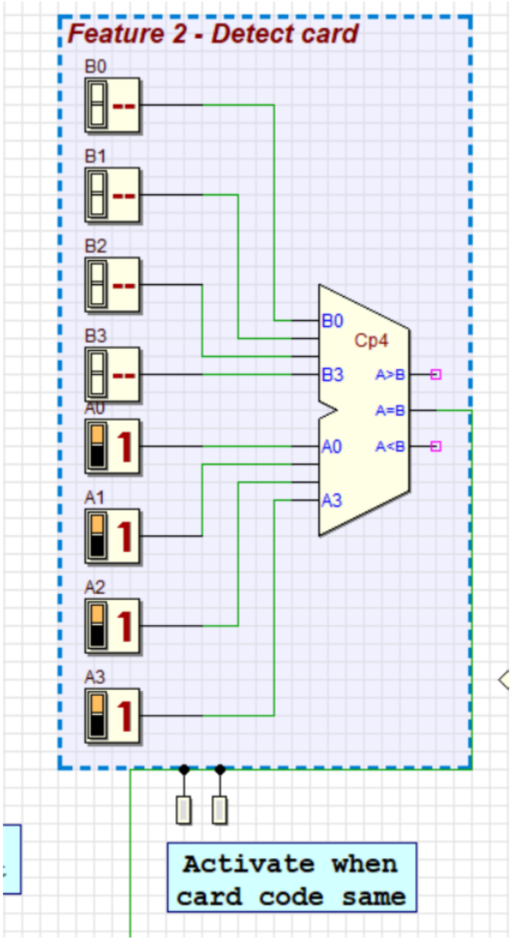
1. Decoder

We want to set a 4-digit password. So, we use 4 bit decoders to decode the set of binary passwords. Each decoder represents each digit of the password. The output of each decoder is connected to two 8 bit LED arrays. It will light up when the user has selected the input value ranging from 0 to F for each decoder. The output of the decoders is connected to two LEDs. Once the LED lights up, it means users have input the correct password and password system is activated. The output of the decoder is also connected to the input of the clock disabler.



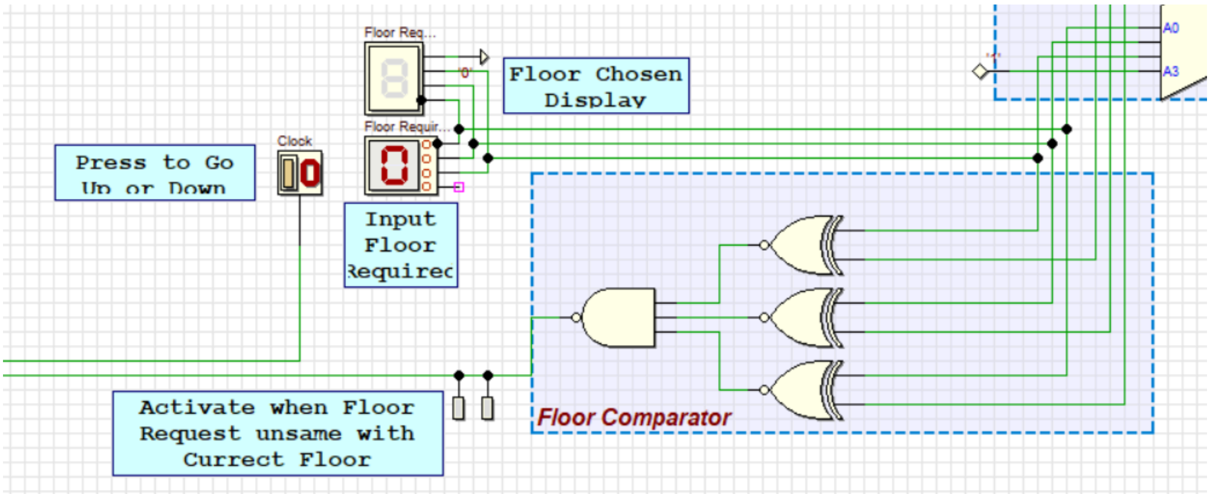
2. 4 bit Comparator

We use a 4 bit comparator to design a system to detect hotel visitor cards. We have set a code (1111) for this system. When residents swipe their cards and the code stored in the card is the same as 1111, the system is activated and two LEDs will light up. The output of the comparator is also connected to the input of the clock disabler.



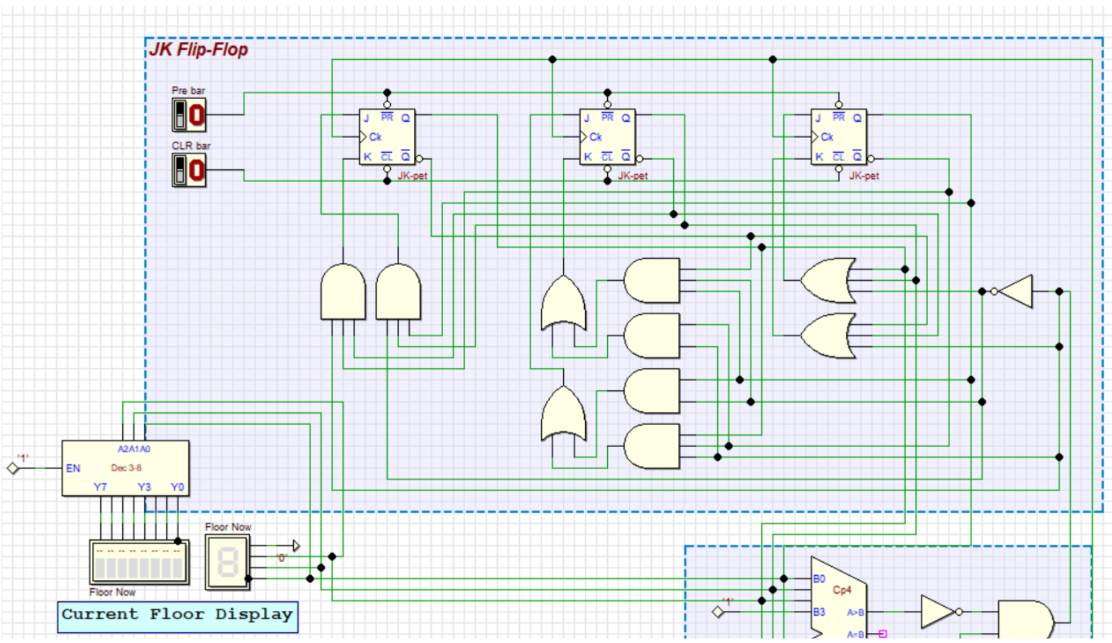
3. Basic gates (3 XNOR gate & 1 NAND gate)

The floor comparator is set up by using basic gates which are 3 XNOR gates and 1 NAND gate. Users have to input the desired floor, when the floor request is not the same as the current floor (comparator is not equal), it sends a signal to the NAND gate to generate output HIGH, LEDs that are connected to the output of the NAND gate will light up. Besides, this floor comparator will produce ACTIVE LOW output and to disable/enable the clock. We also implement a push button instead of a clock generator to eliminate undesirable states during state transition.



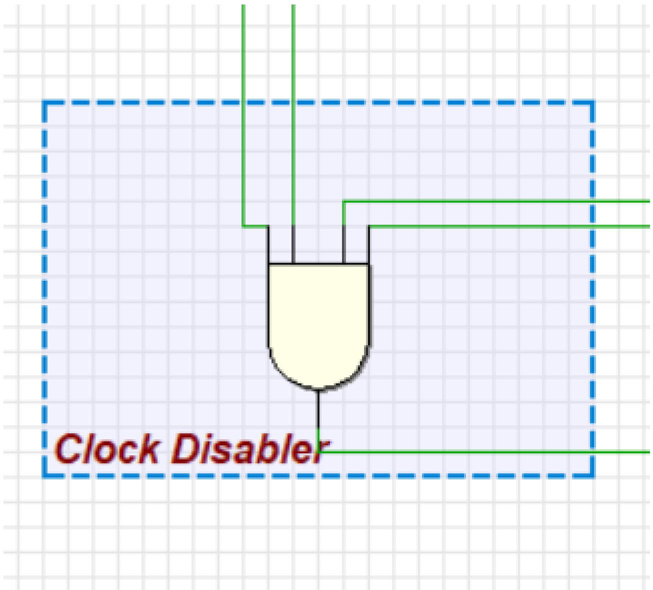
4. 3 bit up and down counter

The 3 bit counter that we use is a 3 bit up JK positive edge count up and down counter. The counter starts or stops to count based on the clock disabler. It will start counting if the values of PRE bar and CLE bar is set to 1 and floor request is not the same as the current floor. It will stop when the clock pulse is no longer received or reaches the number of counts initialised by the user (saturated counter).



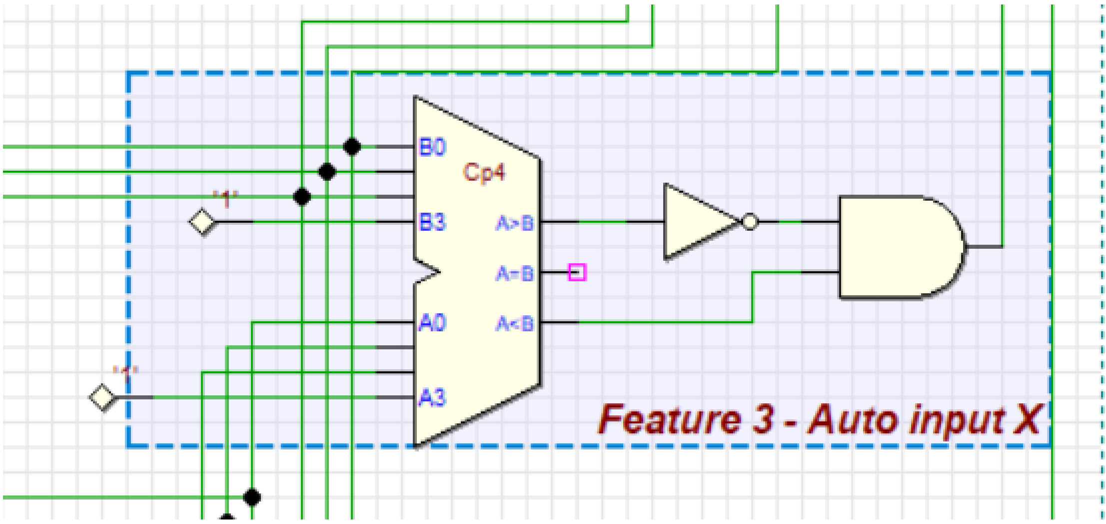
5. Clock disabler

The clock disabler is set up by using a 4-input AND gate. The inputs are signals sent by password system and detect card system (combined using an AND gate), power, clock source and a signal sent by floor comparator. The clock disabler will only activate when all 4 inputs are high. The output of the 4-input AND gate is connected to the clock of the flip-flop of the counter.



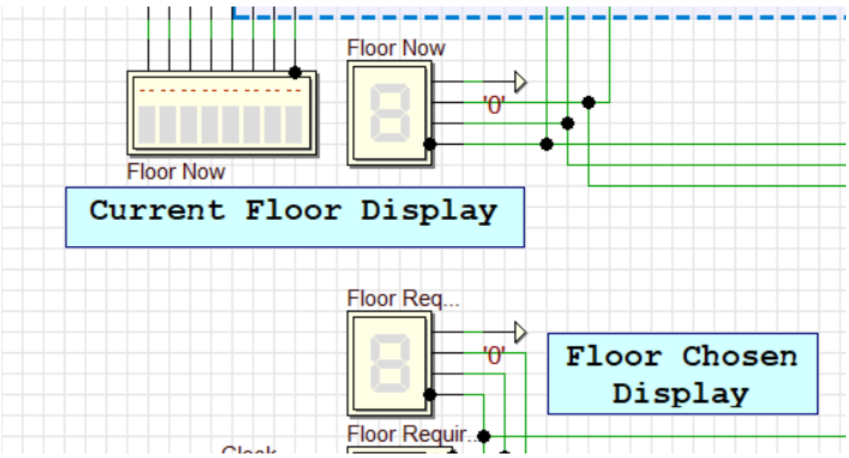
6. Auto input X

We also design a special feature named auto input X by using a 4 bit comparator and an AND gate. The function of this Auto input X is to auto change input X to count up or count down instead of manually change it. It will compare the floor request and current floor to get a signal for count down (1) or count up (0). Element A in comparator is the input floor and element B is the current floor. We add an inverter at output A>B because when the output is A>B, it gives a value 1, but for our counter, value 1 means count down. So, the inverter helps us to convert 1 to 0 to fixed in our circuit.



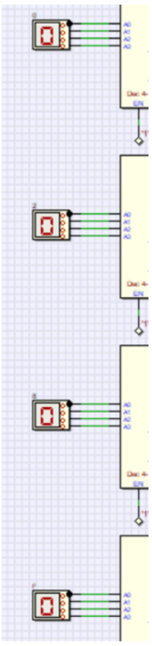
7. 7-segment display

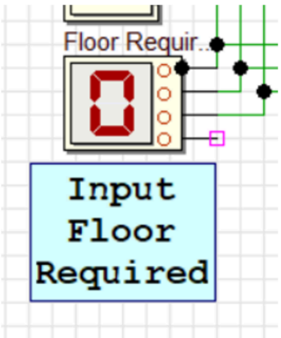
The 7-segment display displays the number based on decimal numbers and letters. It can display decimal numbers 0 to 9 and letters A to F by turning up the proper light. Current floor display and floor chosen display are connected to a 7-segment display.



8. Hexadecimal digit input

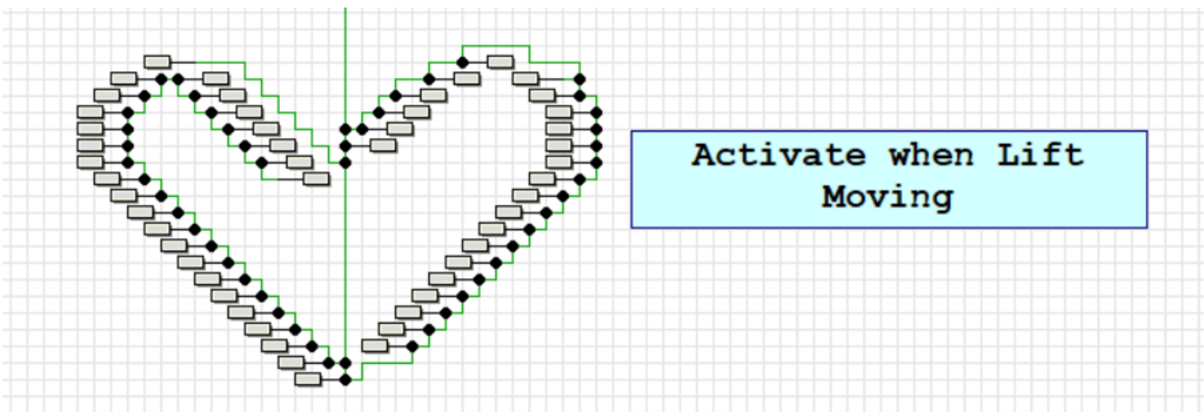
This component allows the user to select input 0-9 and A-F, but we only accept input floor ranges of 0 to 7, as our elevator has floors up to the seventh only. Components that are connected to hex digit input are the password system and input floor required.



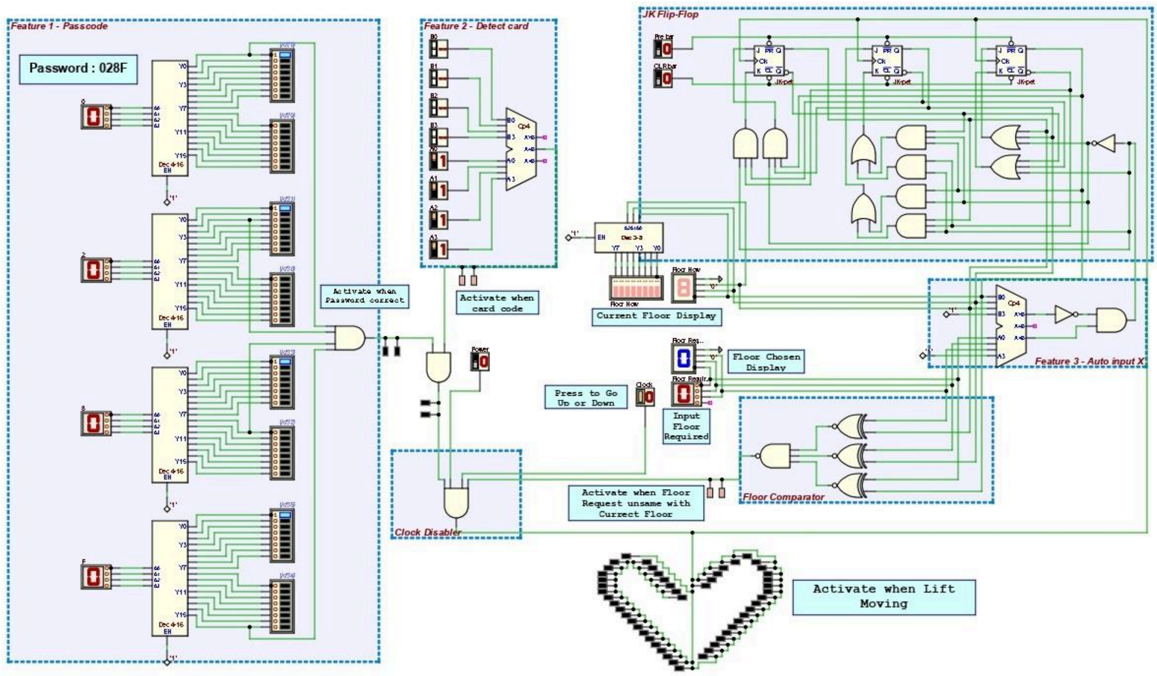


9. LOVE shaped LEDs

We have designed love shaped LEDs to show the moment when lift is moving no matter up or down as it will activate when lift moving. It lights up based on the clock disabler.



Full Deeds Circuit



Conclusion

In conclusion, the development of the electronic controller for the hotel elevator/lift system has been a journey marked by collaborative effort, creative problem-solving, and the practical application of digital logic concepts. As a team, we embarked on this project with the goal of translating theoretical knowledge into a functional solution for the simulated case study.

The user interaction complexity, synchronisation and counting mechanism, integration of JK flip-flops, error handling, and recovery were identified as key challenges in the problem statement. Through meticulous planning and design, we proposed a solution that involves an 8-floor elevator system, utilising a 3-bit up/down counter with a JK flip-flop. The system incorporates additional features such as a 4-digit passcode system, card detection, and auto-input direction to enhance functionality.

The system implementation involved a combination of sequential and combinational circuit components, including decoders, comparators, basic gates, a 3-bit up/down counter, and a clock disabler. The incorporation of these components addressed the challenges posed by the problem statement and enabled the realisation of the proposed solution.

The full deeds circuit showcases the intricacies of the design, including the decoder for the passcode system, the 4-bit comparator for card detection, and various other components contributing to the overall functionality of the elevator controller.

Throughout the project, our team members played designated roles, and the collaborative effort was evident in the distribution of tasks, report preparation, and finalisation. The dedication and guidance provided by Dr. Zuriahati, our lecturer, significantly contributed to the success of the project.

Reflection

The collaborative journey in developing the electronic controller for the hotel elevator/lift system has been a dynamic blend of theoretical knowledge and practical application. Effective team collaboration emerged as a cornerstone for success, with regular discussions fostering a shared understanding and promoting collective problem-solving. This experience highlighted the crucial role of communication and teamwork in project efficiency.

The project served as a tangible link between theory and application, providing a deeper understanding of digital logic in a real-world context. Addressing the complexities outlined in the problem statement, such as user interaction intricacies and synchronisation challenges, sharpened our problem-solving skills and encouraged critical thinking.

Guidance from Dr. Zuriahati played a pivotal role, offering mentorship that clarified complex concepts and inspired creative problem-solving. The implementation of enhanced features, including a 4-digit passcode system and card detection, reflects our commitment to delivering a comprehensive solution that goes beyond basic requirements.

As the project transitions to testing and validation, our focus turns to ensuring the reliability of the implemented electronic controller. This phase serves as a practical culmination of our theoretical knowledge, emphasising the need for rigorous testing to validate performance under diverse scenarios.

In retrospect, this project provided a fulfilling learning experience, bridging academic knowledge with practical application and fostering collaboration within the team. Approaching the final stages, our commitment remains unwavering in delivering a robust and functional electronic controller for the simulated hotel elevator/lift system.

References

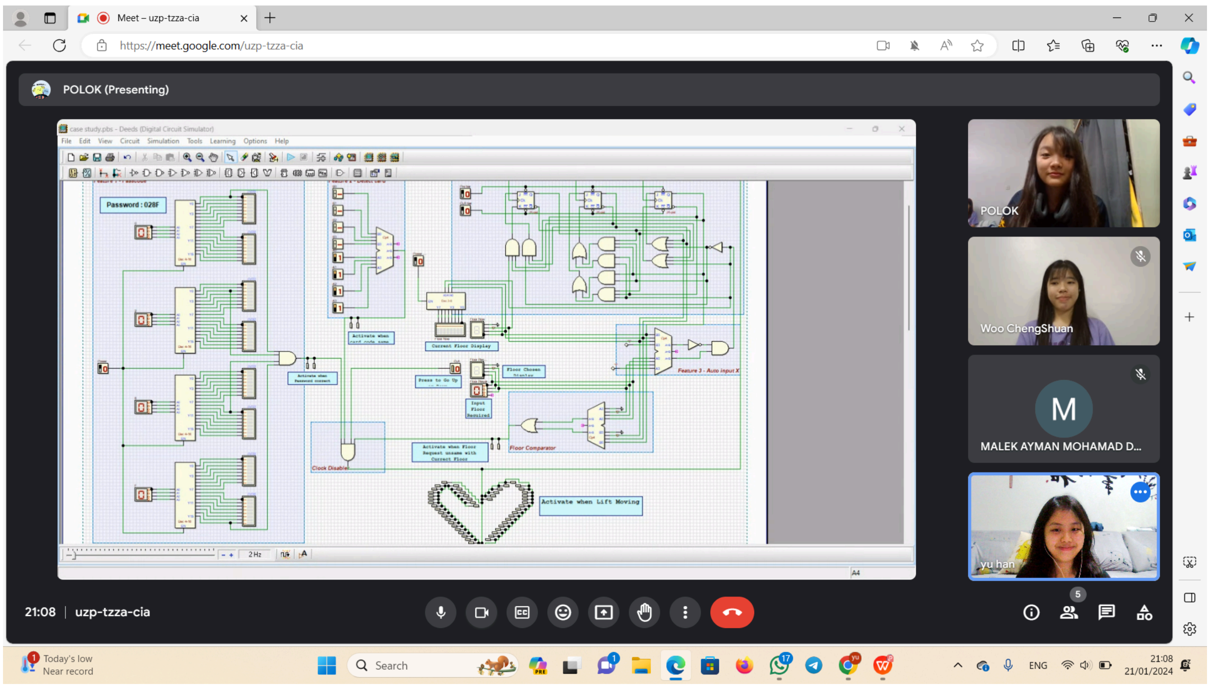
- Digital Logic Modules
- Digital Logic Lab

Appendices

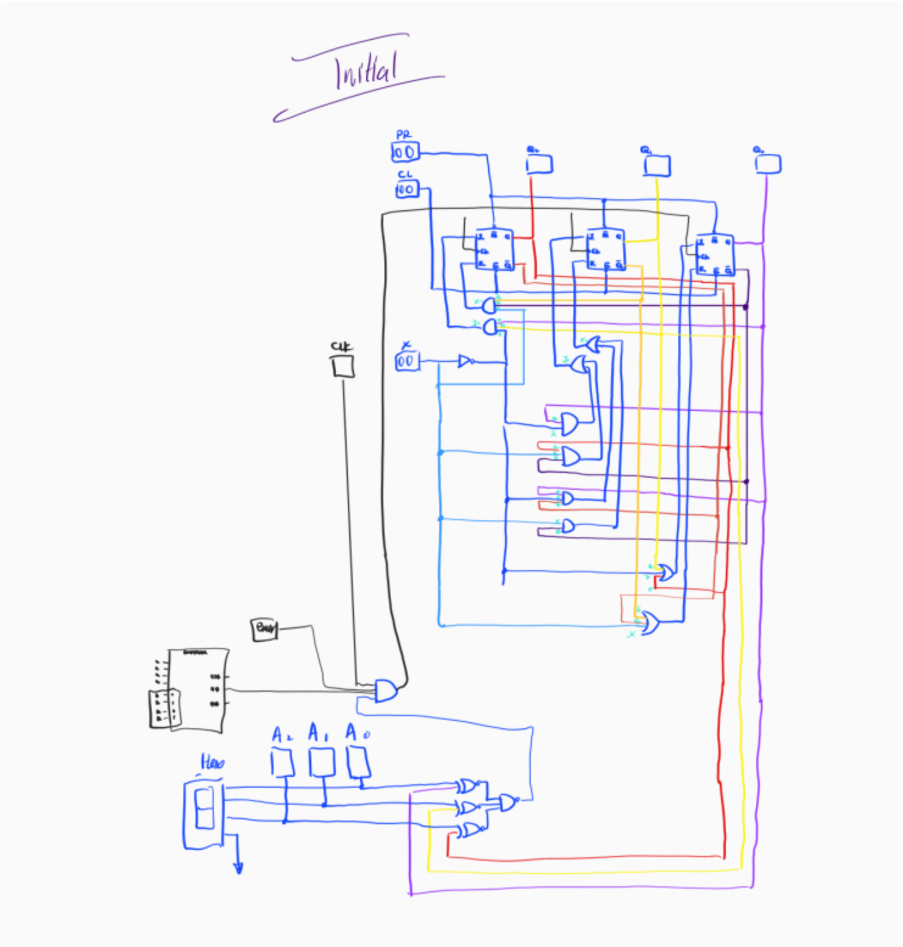
- Task distribution

Group leader	Poh Lok Yee
Deeds designation	Poh Lok Yee, Woo Cheng Shuan
Report preparing	Lim Yu Han , Malek Ayman Mohamad Dahy
Report finalising	Lim Yu Han
Video editing	Woo Cheng Shuan

- Discussion using google meet



- Initial draft



- Final draft